

My Focus was on the Game: Investigating the Use of AI Assistants in a Virtual Reality Exergame for Older Adults

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Figure 1: An older user playing the *Canoe Game* with the AI agent helping them.

Abstract

Intelligent virtual agents (IVAs) show enormous potential to enhance virtual reality (VR) experiences, for example, by providing interactive assistance. In this paper, we provide first insights on the interaction and opinions of older adults communicating with an IVA in a VR exergame. For a user study with eleven older adults, we developed an IVA that was using GPT-4o to naturally communicate game rules and answer questions of the participants. Then we evaluated the interaction between the agent and the older adults. Our findings show that even though the participants enjoyed the communication with the IVA, they were mostly focused on the game and not on the IVA, which was reflected in limited communication between them. Technical issues such as delay or hallucinations further deteriorated the user experience. In this paper, we provide guidelines on how to improve the human-agent interaction between older adults and IVAs in VR exergames.

CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; Virtual reality; • **Social and professional topics** → Seniors.

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Keywords

Virtual Reality, Exergames, Intelligent Virtual Agents, Artificial Intelligence, Older Adults

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1 Introduction

Exergames combine exercising with training and have shown positive effects on healthy aging [13]. In order for exergames to be effective and motivating, they need to show a high usability that is matched to the abilities and preferences of the user [28]. Aside from intuitive interactions [15], a well-designed immersive environment [11], and motivating, meaningful content [9–11, 15], the on-boarding of the game should be designed with a focus on engagement, providing a sense of trust and safety [11]. In this regard, it is important that the goals and objectives of the experience are well-explained and understandable [8]. The user group of ‘older’ adults is heterogenous in itself, with adults in their 60’s having different technological knowledge, experience or cognitive-physical fitness than older adults in their 90’s [7], and the information should be provided in a suitable way. Especially for first-time VR users, controls of the application, rules, or safety notices should be explained and communicated in an easily understandable way. In this

regard, reading a lengthy instruction sheet and the insecurities of (new) users with the technology can be a usage barrier.

One approach to solve these issues is through intelligent virtual agents (IVAs). These are computer programs that mimic human communication and social interaction [6]. IVAs can range from text-based chatbots to voice assistants, and to embodied agents. In prior research with older adults, IVAs were used to practise patient-doctor conversations [20], or to show older adults how to perform easy balance exercises [21]. IVAs can be represented in a variety of shapes. Albaina et al. evaluated Flowie, a virtual coach in the shape of a flower that encourages older adults to walk more [1]. In one of our prior works, a humanoid agent, an abstract text/voice representation and a virtual parrot were compared [19]. Here, we found that younger and older adults preferred the parrot, but also that the personality of the user has an effect on their perception of the agent.

AI has made huge advancements in the last years [23, 27], with the ability of LLMs like the generative pre-trained transformer 4 (GPT-4) to produce human-like text output based on the user input it receives [22], starting a new era for communicating with IVAs. With LLMs, natural conversations with IVAs are made possible. In VR, a GPT model was previously integrated as an assistant in an escape game, which was evaluated with younger adults [24]. Here, it helped the players to solve riddles, but limitations of the model regarding its knowledge of VR-specific contexts and several vague responses were observed.

When talking about AI and IVAs, most of the time, younger adults are asked about their opinion, excluding the view of older adults in using the technology. Nevertheless, this user group has their own opinions on this topic, with their own barriers, but also their own potentials for using it to extend their daily life, and it is important to ensure a more inclusive and comprehensive understanding of how these technologies can be used across all age groups. With this work, we aimed to get a first glimpse into using IVAs in VR exergames played by older adults. Our aim was to see how an IVA that is communicating verbally using GPT-4o and text-to-speech (TTS) is perceived by the older adults. Furthermore, we were interested to analyze the conversations the older adults had with the IVA, identifying potentials, but also errors of the IVA that lead to misunderstandings. With this knowledge we provide guidelines on designing IVAs using LLMs in exergames for older adults.

2 Methods

As an application, we used the *Canoe Game*, which has been evaluated with older adults for a variety of different research questions [14, 16, 19]. In this VR exergame, users sit in a virtual canoe that runs down a river all while they have to touch soap bubbles with their hands, according to different rules. For this paper, we extended the game by including an IVA that explains the game rules, answers questions of the participants, and provides feedback on the game performance. A flow diagram of the game is shown in Figure 2, detailing the different steps, as well as the AI agent's prompts.

The game starts in augmented reality (AR) passthrough mode allowing users to seamlessly transition into the virtual environment.

This helps them adjust their vision and become familiar with the VR controllers, ensuring a smooth and easy on-boarding experience. When users feel comfortable, a button press starts a transition to the VR experience, allowing them to take control over the game flow [17]. They then have to calibrate their sitting position by pressing and holding both trigger buttons for three seconds, followed by a calibration of their range of motion, which is done by pushing four balloons as far away as possible. After this is done, they can press a button on their right side to start the canoe, and subsequently collect the soap bubbles. At the end of the game, a scoreboard shows them how many correct bubbles they collected.

To integrate the IVA, we added an animated 3D parrot (See Figure 1) to fit to the tropical island scenario [19]. The parrot uses the OpenAI model *GPT-4o* with the OpenAI Text-to-Speech (TTS) voice "Fable" for speech generation in German. Users are able to communicate with the agent naturally through their own voice. The spoken words are then transcribed using the Google Speech-to-Text API and sent to GPT-4o as a user prompt. Voice input is only recognized while the IVA is not talking itself, avoiding interference caused by concurrent voice input. A sign in the virtual world indicates to the participants when they are allowed to talk (a mouth, with a green border), or when they can only listen to the parrot (an ear, with a red border). This was explained and shown to them before the study started.

For the technical implementation, a system prompt details the expected interaction and medium of communication, ensuring the agent's answers can easily be transformed to audio using TTS (See appendix). The prompt contains contextual information about the game in the form of the scene, task, expected events and instructions the agent should communicate to the participant as a step-by-step guide. After each sequence of the game is completed, e.g. the calibration or the pressing of the start button, the actions performed are added to the chat history in the shape of a written description, including the contents of the next expected response. For example, if the calibration is successfully finished, this is communicated to the agent by sending a prompt stating that the user is done, and the next step can begin. The agent cannot directly access the state of the game, e.g., verify whether what is communicated is true – at this point, it is only text-based and the agent relies on truth being communicated. A translation of the exact prompts given to the agent in each situation can be found in the appendix.

2.1 Study Procedure

We conducted the study in cooperation with a local senior living home on their premises. Upon arrival, participants signed a consent form and received a small briefing about the goal of the study. We did not tell them anything about the VR game itself, but rather explained that they could ask the virtual agent what the game was about, and how to play it. We also gave them a presentation of how to talk to the agent, showing them the visualizations of the ear for listening and the mouth for talking.

For the study, we used a Meta Quest 3 HMD with two controllers. It was connected to a 4G+ mobile network. The participants then put on the HMD and started in the passthrough mode, where they had time to adjust the HMD and controllers. The parrot introduced itself and asked the participant if they had any questions. Then, it



Figure 2: Gameflow (red nodes) as well as input of the IVA provided in different situations (blue nodes).

continued to guide them through the calibration and the game flow. After each level, the participants could decide to either continue playing, or end the game. At the end of the game when the scoreboard was shown, the IVA congratulated the player, informed them how many bubbles they collected and told them they could either take the HMD off, or continue the conversation with the parrot.

To assess how well the game and the IVA were received, we then performed a semi-structured interview with the participants, where we asked them about their experience with the IVA, their wishes, fears and if they could imagine using an AI agent in real life. The interview questions can be found in the appendix. Additionally, we logged the prompts that the participants sent to the IVA, as well as the agent’s responses, and took observation notes during the interaction.

3 Results

Eleven older adults participated in the study. All of them were able to give legal consent and did not have a diagnosed cognitive impairment. They were between 77 and 91 years old, with an average of 84.54 ± 3.73 years. Nine identified as female and two as male. Two of them had used an AI or voice assistant on a computer before. Seven participants had used a VR headset in the scope of user studies before, and the other four had experience with an immersive projector-based gaming system called the ExerCube [26].

3.1 Interview Evaluation

To evaluate the semi-structured interviews, we first transcribed them and then performed an inductive thematic analysis on the

data to find common themes [3–5]. In the following, we report the main findings, which relate to the themes ‘Communication’, ‘Focus’, and ‘Possibilities of AI’. We denote P_{1-11} for the individual participants.

Communication. Most of the participants rated the communication as very positive. Participants found that the IVA was patient (P_{11}), understandable (P_4, P_5, P_9), and appreciated that instead of trying out which buttons work, they had someone to ask their questions (“If someone is there, one can ask”, P_{11}). Nevertheless, some participants also encountered technical issues in the communication. For example, at some point, the agent did not respond anymore or a reply from GPT took too long to get through (P_2). Also, for one participant, the agent kept repeating sentences, which was perceived negatively (P_7).

Focus. Many participants reported that their focus was not on the IVA, but on the game, even though the briefing informed them about the goal of the study. P_1 reported that they did not talk much to the agent, because they were just focusing on the game. P_{10} also stated that during the game, their attention was on the task and if the agent had talked, this would have bothered them. This was also confirmed by P_6 , who paid little attention to the possibilities of talking to the IVA, but rather followed the instructions of what the parrot explained. They were focused on something new and were not interested to find out facts like, for example, why the game was developed.

Possibilities of AI. Before the study started, the possibilities of asking the agent questions were explained to the participants, e.g.,

that they could ask questions about the game, but most older adults did not use this feature. In the interview, after being reminded of this possibility, P_1 mentioned that they should have asked the agent something. After giving P_3 a specific example, they said that if they had known this before, they could have used the AI in a different way, but they simply did not know. P_{10} stated that if they had more insight into the topic of artificial intelligence, it would be easier for them to deal with it, but they were not interested to invest too much time in learning about it. P_8 also said that they did not know which other possibilities there were, and instead of getting a list of all possibilities, they would let themselves be surprised. Finally, when asked if they would play the game without the researcher present, and only with the parrot, all eleven participants confirmed that they would.

3.2 Interaction Evaluation

During the study, all prompts were recorded, along with observations from the leading researcher. On average, each user sent 5 ± 2.17 prompts to the agent with a total of 55 prompts over the eleven participants. The evaluation of the conversation content through Thematic Analysis resulted in two interesting topics: ‘*Misunderstandings and Hallucinations*’, and ‘*Timing and Initiative*’, which will be presented in the following paragraphs.

Misunderstandings and Hallucinations. Out of the total of 55 prompts, 15 were statements like “okay”, “ready”, “done”, or “thank you”. From observations, prompts like “done” sometimes lead to misunderstandings, resulting in the IVA continuing to explain the next game rule, even if tasks were not successfully finished yet. For example, this happened if the calibration button was only touched for a short time, or if not all of the balloons were touched yet. The participants believed they were done, communicated this, and this false belief in turn was transferred to the IVA (P_6 : “*Was this enough?*”, IVA: “*Yes, this should be enough. Lets continue with the second step*”).

Another twelve prompts were questions about the game, for example which button to press (P_6), how to reach the bubbles (P_{11}), what comes next (P_9) or the inquiry to repeat what the agent just said (P_2). Further, they asked for confirmation on if they should perform an action now (P_2, P_7, P_{11}). Here, another limitation became visible - hallucinations. This happened especially when participants asked for a confirmation of the correctness of a planned action, where the agent tended to always agree with the user. For example, P_6 asked if “*the black round start button*” was the correct one to start the game, whereupon the IVA confirmed it: “*Yes, exactly. Press the round black start button to start the canoe*”. In this case, the participant was looking at the trigger button on the controller and the correct button was spatially on their right side and colored in red. A similar error happened for P_2 , who commented that she had “*a finger back [on the trigger button] but nothing [was] moving*”. The IVA then explained that the start button was on the right side of the screen and that they should point the controller onto the button and then press the trigger button. Again, this was not correct, since the button worked by simply pushing it away with the controller, similar to a real-world button.

Timing and Initiative. Most of the questions occurred during the calibration of the game. Another eleven prompts followed immediately after the initial greeting, where the users were asked if they had any questions about the game. Here, only two participants asked a question regarding what the game was about (P_1), and what the IVA was planning to do with them (P_8). All others stated that they currently did not have any questions. During the gameplay, the participants did not ask questions anymore, but rather voiced some comments about the gameplay, for example that the bubbles were “*too high*” (P_5), “*I can’t do this*” (P_{11}), or reminding themselves of the game rule “*No, that’s left, red right, left [yellow]*” (P_5). After the game ended, the parrot told the users that if they wanted, they could continue talking to him. Similarly to the start, only two participants used this opportunity. One user asked the parrot how many bubbles they missed (P_9), while another one told him that the topic of AI was very unfamiliar to them (P_{10}), which was followed by a short explanation of AI by the IVA. The other users ended the game or took the HMD off. Another interesting observation was that often, participants made understanding comments while the IVA was talking. In order to prevent interference, these comments or interruptions were not recorded at this point, because the IVA only listened for input while it was not talking, but for future implementations, this could foster a more natural conversation and signal understanding of the participants.

4 Discussion

In this section, the results will be discussed and put into context. Furthermore, guidelines for future research are presented.

4.1 Qualitative Discussion

While the qualitative evaluation showed great potential for the use of IVAs, e.g., having the opportunity to ask questions, the main focus of the participants was on the game itself, and not on the IVA – similarly to a cognitive training study conducted with an IVA, where the attention of the users was not on the agent either, but on their task [18]. In our case, the IVA guided the conversation and explained the next step automatically when an action, for example the calibration, was completed. In future research, it would be interesting to see how the conversation would be shaped if this behavior was different, e.g., if the participants had to actively inquire about each step to receive more information, or in more collaborative scenarios. Nevertheless, we suspect that this might decrease the perceived efficiency of the game. After all, the IVA only served as a helper and, for the older adults, was not the main part of the experience. As researchers, we were interested in the results of a user study, however, the older adults were most likely more interested in experiencing the game content and benefiting from the immersive training. Therefore, these fast-paced exergames might not be the perfect use case to evaluate the integration of IVAs.

On a positive note, we can confirm that all players were able to navigate through the game without input from the researchers. This is also confirmed by the interview data that stated that all players would be happy to play only with the IVA, without a researcher present. Nevertheless, it remains an open research question whether, in reality, this is feasible and also safe. Especially for movement-intensive games, we believe that there should be a human observer

present to intervene in case of a (medical) emergency, at least until AI capabilities are advanced enough to ensure the safety of the older adults at all times.

4.2 Technical Discussion

The largest technical limitation we encountered during the study was the latency in the responses from the agent, which resulted from the slow internet connection we experienced while at the facility. This influenced all the connections for interacting with the agent, namely STT, AI and TTS. While the connection speed is the main cause, in each of these steps, improvements are possible and necessary to result in smooth, natural communication.

Further, we encountered several hallucinations of the agent. While LLMs are prone to hallucinate unintended outcomes, there are different ways to prevent this. First, we only observed extrinsic hallucinations, which are hallucinations that cannot be verified from the source input [12]. In our case, this was observed with the color of the button (original: red, hallucination: black), which we did not include in our initial prompt to the agent, and the interaction with the button (original: pressing, hallucination: raycasting). In future implementations, more thorough pre-testing might be able to identify these issues and prompts can be adjusted. Another idea would be to mitigate the hallucinations through modeling and interference methods [12], such as reinforced learning or fine-tuning. The IVA also tended to agree with the user when they asked a question, without having the corresponding information. Rychert et al. integrated a prompt into their agent that stated to “*never assure the user that they can do something with a specific object*” [24], which might also help to decrease the certainty with which the agent suggests actions in our implementation. If the agent does not have information about a certain topic, it could also visually and audibly show signs of uncertainty, as it was also done in prior research [25]. Further, the latest advancements of OpenAI’s GPT feature vision abilities. Including this could help to make the intentions of the users more clear, e.g., by detecting which objects their gaze or attention is on, and could also account for the imprecise speech of the users, e.g., “*Should I press this button?*”. We also suggest to include state awareness, e.g., by always sending the state of the game and of interactable objects to the IVA in each prompt [24]. This could help to reduce errors from users falsely claiming to be done with a task.

4.3 Guidelines

With the knowledge gained from our exploratory user study, we propose the following guidelines for designers and programmers planning to integrate IVAs into their applications that are made for older adults.

1. *Make the Agent More Approachable.* Often, the older adults were not aware of all the possibilities the IVA provided, or they had hesitations to ask questions. One idea would be to make the agent more approachable, e.g. with a friendly introduction, emphasizing that it is in the game to help the players and that they can always ask questions. Further, it could give examples on what to ask. This could, on one hand, lower the barrier to ask questions in the first place, and on the other hand, it would help players to discover more of the ways they can use the IVA.

2. *Provide Enough Knowledge to the IVA System.* Prompts and information given to the AI should be as specific as possible, with as much information as possible. Even through this information is not new, the specific interaction patterns of older adults might differ from younger users, e.g., through their reduced experience with technology. While we aimed to provide all information necessary for the gameplay, which was even more than prior versions of the game communicated, we could still observe hallucinations where the IVA filled in missing knowledge with suggestions of the user, e.g., touching the wrong button. Through careful playtesting with the end users in a user-centered design approach, common misconceptions can be identified and prompts can be altered. Further, in our implementation, the whole conversation was guided through user prompts. Here, the LLM did not make a difference between prompts that were sent by our system, and prompts that originated from the verbal conversation with the older adults. This led to misunderstandings if the participants falsely communicated the successful execution of an action. We, therefore, propose to implement state awareness in the IVA, e.g., through the ability to access the state of the game flow [24], or by integrating new advancements such as vision abilities.

3. *Consider Whether to Really Use AI.* While using AI has many advantages, there might be situations where predefined dialogues surpass the non-deterministic communication of IVAs. Posing the same question several times can lead to different outcomes, and some of those might not be the desired ones, e.g., wrong or misleading information. Designers should carefully consider whether there are situations where exact words are needed to understand the instructions or context – which could be implemented without the integration of AI. Furthermore, environmental and ethical considerations of using LLMs [2] should be kept in mind when using them in large-scale implementations.

4.4 Future Work

Our study design shows several limitations that should be addressed in future work. First, we only evaluated one condition, namely the exergame with the integration of an IVA using GPT-4o. This was done to get a first glimpse into the usage of IVAs by older adults, and to inform us about possible barriers and difficulties. In future studies, within-subject studies should be conducted to compare different implementations with each other, e.g., an LLM-based IVA with a pre-scripted dialogue system. Further, technical limitations such as a long delay should be minimized, or ways to bridge the waiting time should be implemented. Moving on with this topic, we plan to evaluate how prior knowledge of using LLMs might benefit the interaction between older adults and the IVA. Furthermore, we plan to investigate the interaction between IVAs and older adults in different use case scenarios, e.g., for physical or cognitive therapy. It would be interesting to investigate the interaction between older adults and AI assistants in more collaborative and communicative scenarios, e.g., when solving riddles together [24, 29].

5 Conclusion

In our exploratory user study, we evaluated the feasibility of an IVA in a VR exergame that assisted the players and guided them through the gameflow. In interviews, we found that the communication with

the IVA was perceived as positive, the participants' questions could be answered understandably and all of them stated that they would also play the game alone with the IVA, without a researcher being present, fostering independence.

Nevertheless, technical limitations such as a high delay and hallucinations deteriorated the experience. Especially wrong information being communicated by both, the IVA and the users, led to misunderstandings. State awareness is needed to reduce these occurrences. Further, we found that the older adults in our study rather focused on the fast-paced game, and the conversation with the IVA only played a subordinate role. We propose that the possibilities of AI should be explained and demonstrated to the older adults by the IVA, informing them about possible use cases. Further, the necessity of implementing AI-based agents (as opposed to scripted dialogues) should be evaluated depending on the use case.

While it is easier to use pre-scripted tutorials, IVAs present a possibility for further tailoring the agent's behavior and communication to the needs of the individual users. In the future, this might enable older adults to access technology such as VR exergames more independently, empowering them to ask questions and fully enjoy the benefits these tailored training programs offer.

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A GPT Prompts

This section lists the exact prompts given to GPT during the game. These have been translated from German to English for this paper, using ChatGPT.

A.1 General Information

The following prompt was provided to the IVA at the start of the game. It contains all information on the game, the game flow as well as the game rules. We found it beneficial to provide this information right from the start in order to empower the IVA to answer questions and be aware of the game flow before reaching certain events:

You are a parrot named Ara, controlled by artificial intelligence, in a virtual reality game programmed by the University of Hamburg. Your task is to assist older adults by providing them with tips on interactions and the game rules. Please address me formally and use polite language.

Importantly, the text will be converted to speech, so avoid complex punctuation like asterisks, semicolons, or similar. Also, try to keep explanations fluent and as brief as possible. Use simple words that older adults can easily understand. Speak in German and refrain from using foreign words, English terms, or slang. Do not ask after every sentence if you can assist further.

I will explain to you the rules and setting of the game. I must go through the steps in the given sequence. Explain one step at a time to me and do not jump ahead to future steps. If I ask you questions in between, answer them in an easily understandable manner.

The game takes place on a tropical island. I am wearing a virtual reality headset, the Meta Quest. With it, I perceive the world around me in stereoscopic 3D. I have a controller in each hand, which are small handy technical devices with various buttons. They have trigger buttons. These are located at the back of the controller and can be reached with the index finger.

Step 1: The game begins with a calibration of my seating position. For this, I have to look forward and press the two trigger buttons on the controllers for three seconds.

Step 2: Next, my range of motion is measured with four balloons. These balloons are located above and beside me. If they can't be found, I should look up a bit. The balloons are very sluggish. It's enough to briefly touch them. When I touch them, they move further away from me. If I'm satisfied with the distance, I can simply lower my hands briefly. The blue balloons must be touched with the left hand and the orange balloons with the right hand.

Step 3: Then the real game starts. The game involves paddling down a river in a virtual canoe. Soap bubbles appear in front of me, which can be popped. There are three levels in total, each lasting one minute. There are different rules in the levels. In the first level, all soap bubbles should be destroyed. In the second and third levels, various mental tasks are added.

The game is intended to keep older adults like me fit in advanced age. Targeted movements and mental tasks strengthen my abilities. The game should not be too strenuous. It is not a test, but should be fun. It is played while sitting. The game automatically becomes more difficult if many correct bubbles are popped and automatically becomes easier if I make some mistakes.

The rule of the first level is to touch all soap bubbles. The rule in the second level is to pop all red soap bubbles with the right hand and all yellow soap bubbles with the left hand. The rule in the last level is

to alternate between popping a bubble with a letter above it and then a bubble with a number above it.

A.2 Situational Prompts

At certain points in the game, additional prompts were given to the agent, independent of the verbal input of the user. These prompts informed the agent about the current state of the game and gave further instructions on how to proceed. Prompts were given after transitioning from passthrough to the immersive environment, after a successful calibration of the sitting position, after calibration of the ROM, once a level was started, and once it ended. Further, the agent made comments if the user played well, or if they made mistakes. The specific prompts are the following:

Introduction: Introduce yourself with your name Ara and say that you are an artificial parrot. Then ask if I have any questions about the game.

After calibration: Finished with the calibration of the seating position. Now explain to me that my range of motion will be measured. For this, I need to push four balloons as far away from me as possible.

After ROM calibration: Finished with the measurement of the range of movement. Now there is a start button on the right side. As soon as it is pressed, the canoe will slowly start moving. Point out to me that I should immediately inform you if I start feeling seasick. If I do, it helps to close my eyes and quickly take off the headset.

After pressing the start button to play a new level: Explain to me the rule for level number [1,2,3].

If the user was doing well: Give me a compliment because I'm playing so well.

If the user is making mistakes: Politely point out to me that I'm still making some mistakes and explain the rule of the level again.

At the end of a level: The level is over. Ask me if I want to play again or if I want to end the game. The corresponding buttons are on the right side.

At the end of the game: "The game is now over. Congratulate me and tell me that I collected [amount] correct soap bubbles. Point out to me that I can take off the virtual reality headset now, or I can continue talking with you if I want.

B Semi-Structured Interview Questions

The following questions were used as a basis for the semi-structured interviews:

- (1) How did you find entering the virtual world? Did you like the passthrough mode?
- (2) Were you able to hear everything clearly?
- (3) Was there anything you didn't like? Do you have any wishes?
- (4) How did you find it to play with the parrot as a helper?
- (5) Did you miss having a human contact person? Would you also play the game without the researcher present?
- (6) How intelligent did you find the parrot? Did it make mistakes?
- (7) What else would you have wished for from the parrot?
- (8) What behaviors should it show more of? What should it stop doing?
- (9) Have you used AI before, such as GPT? What do you think about it?
- (10) Can you imagine using AI in real life as well? If so, for what and how?